

4.4 SUPER CLASS: AGNATHA (Gr. A- WITHOUT; GNATHOS- JAW)

They are jawless primitive fish like aquatic vertebrates without paired appendages. The internal ear bears one or two semicircular canals. Notochord persists in the adult.

Agnatha includes the extant class **Cyclostomata** besides the extinct class **Ostracodermi**.

4.4.1 Class: Cyclostomata (Gr. *cyklos*- circular; *stoma*- mouth)

Cyclostomes are jawless aquatic (marine or freshwater) forms. Body is scaleless, long, slender and eel-like. Endoskeleton is cartilaginous. Vertebrae are represented by imperfect neural arches in some. Mouth is circular and suctional. Tongue bears **horny teeth**. Respiratory organs include six to fifteen pairs of gill slits. Heart is **two-chambered** as in fishes. **Renal portal system is absent**. Kidneys are **mesonephric**.

e.g. *Petromyzon* (lamprey) and *Myxine* (hag fish/slime eel).

Marine **lampreys** migrate to freshwater for spawning (anadromous migration). After spawning they die. Their larvae are **ammocoete** metamorphose into adults and return to the sea.

Hagfishes are exclusively marine. They are considered scavengers because they feed on dead or dying fish. When disturbed hag fish releases a milky fluid from its mucous sacs which becomes slime on contact with sea water. Development is direct.



Fig. 4.6 *Petromyzon*

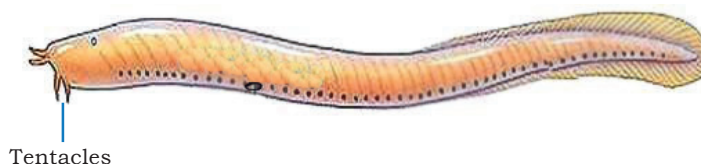


Fig. 4.7 *Myxine*

4.5 SUPER CLASS: GNATHOSTOMATA (Gr. GNATHOS- JAW; STOMA- MOUTH)

They are vertebrates with jaws. They possess paired appendages (pectoral and pelvic fins or fore and hind limbs) and internal ears with three semicircular canals each. Notochord is replaced partly or wholly by the vertebral column.

Gnathostomata includes **jawed fishes** (Pisces) and **Tetrapods** (Amphibia, Reptilia, Aves and Mammalia).

- Pisces and Amphibia constitute the group **Ichthyopsida**
- Reptilia and Aves constitute the group **Sauropsida**.

4.5.1 Pisces (L. *piscis*- fish)

Fishes, the 'first jawed vertebrates', are completely aquatic and gill breathing animals. The extinct jawless fishes, the **ostracoderms** gave rise to the earliest jawed fishes during the **Silurian** period. **Devonian** period is considered the '**Golden age of fishes**', because they emerged as the predominant group during that time. They constitute the most **diverse** and **largest** vertebrate group. They show variation in their body size, shape, and colour. They are one of the economically most important animal groups to mankind. **Coelacanth**s (marine) and **Dipnoi** (fresh water) are the extant 'lobe finned' fishes. **Sharks**,



Fig. 4.8 *Latimeria chalumnae*

rays and **ray-finned fishes** account for the bulk of living fishes. Osteolepid fishes which are of the same group as *Latimeria* are believed to be in the line of evolution of the next higher group of vertebrates, the amphibians. The study of fishes is known as **Ichthyology**.

Name the fish which was thought to be extinct until 1938, but later found to be living in small numbers.

General characters

- ❖ Fishes are completely aquatic poikilothermic (cold blooded) animals.
- ❖ Body of a fish is usually streamlined and differentiated into head, trunk and tail.
- ❖ Exoskeleton consists of **mesodermal** scales or bony plates. A few are scaleless.
- ❖ Endoskeleton may be cartilaginous or bony. Skull is **monocondylic**. Vertebrae are **amphicoelous** (centrum is concave at both anterior and posterior faces).
- ❖ Locomotion is assisted by unpaired (median and caudal) fins along with paired (pectoral and pelvic) fins.
- ❖ Mouth is **ventral** or **terminal**. Teeth are usually **acrodont**, **homodont** and **polyphyodont**.
- ❖ Exchange of respiratory gases is performed by the **gills**. Heart is 'two chambered' and is described as **branchial heart** as it supplies blood only to the gills. The circulation is single circulation as blood reaches the heart only once, in the course of each circulation, making the heart a **venous heart** (heart receives only venous/deoxygenated blood from the body parts).

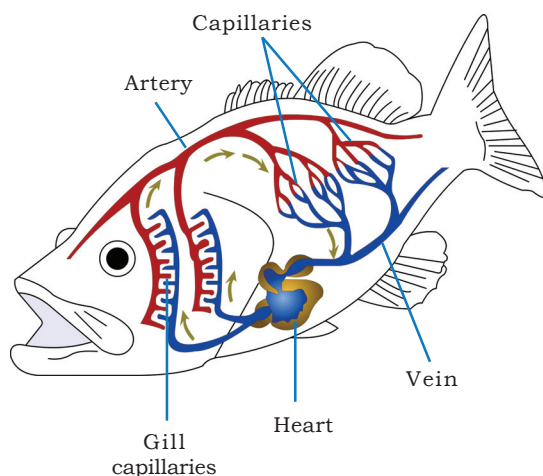


Fig. 4.9 Circulation of blood in a fish- schematic representation

- ❖ Kidneys are **mesonephric**. Fishes are mostly **ammonotelic** and some are **ureotelic** (cartilaginous fishes).
- ❖ Cranial nerves are 10 pairs. **Meninx primitiva** is the only 'meninx' enveloping the central nervous system.
- ❖ Internal ear consists of three semicircular canals. **Lateral-line sensory system** (to detect movement and vibration in the surrounding water) is well-developed. Eyes are without eyelids and each eye ball is protected by a **nictitating membrane**.
- ❖ Sexes are separate. Fertilization is internal or external. Development may be direct or indirect.

Pisces include two extant classes i.e. **Chondrichthyes** and **Osteichthyes**.

4.5.2 Class - Chondrichthyes / Cartilaginous fishes)

(Gr. **chondros**- cartilage; **ichthys**- fish)

They are marine fishes which possess cartilaginous endoskeleton. Caudal fin is **heterocercal** (asymmetrical both externally and internally). Skin is covered by dermal **placoid** scales also called dermal denticles. Sharks are highly predaceous. Respiratory gaseous exchange is performed by five to seven **lamelliform** gills, without operculum, on each side. Air bladder is absent, and so they have to swim constantly to avoid sinking (bony fishes have 'air bladder' acting as a 'hydrostatic organ' helping the fish float easily at the desired level without much expenditure of energy). Cartilaginous fishes are **ureotelic** and store urea in their blood to maintain osmotic concentration of body fluids (physiological uraemia). In males, pelvic fins bear **claspers** to facilitate internal fertilization. They are mostly **viviparous**.

e.g. *Scoliodon* (dog fish), *Pristis* (saw fish), *Carcharodon* (great white shark), *Dasyatis* / *Trygon* (sting ray; possesses poison sting), *Torpedo* (electric ray; dorsal muscles are modified into electric organs), *Sphyrna* (hammer - headed shark).

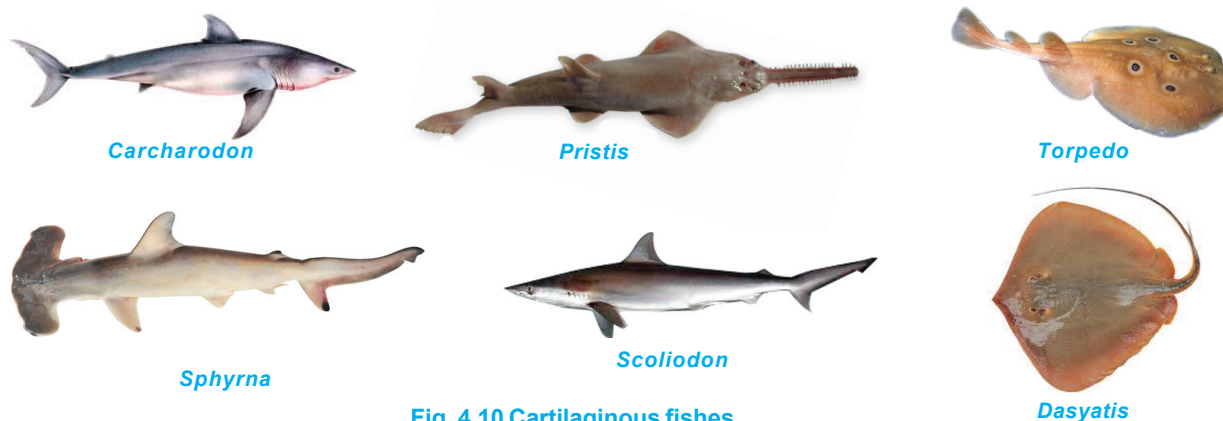


Fig. 4.10 Cartilaginous fishes

Why are sharks ureotelic, while most of the bony fishes are ammonotelic?

4.5.3 Class - Osteichthyes (Gr. *osteon*- bone; *ichthys*- fish)(Bony fishes)

They are bony fishes which live in all kinds of aquatic habitats. Caudal fin is **homocercal** (symmetrical externally and asymmetrical internally) in **teleost fishes** and **diphycercal** (symmetrical both externally and internally) in lung fishes and *Latimeria*. Exoskeleton is in the form of **cosmoid**, **ganoid**, **cycloid** or **ctenoid scales**. Mouth is usually **terminal**. Respiratory gaseous exchange is performed by four **filamentous gills** covered by an **operculum**, on each side. An air bladder is present with or without connection to the gut. It is either helpful in gas exchange (lung fishes) or in maintaining buoyancy (hydrostatic function) in most of the 'ray-finned' fishes. They are mostly **ammonotelic**. Sexes are separate. Fertilization is generally external. Most forms are **oviparous**.

Examples: Marine fishes: *Exocoetus* (flying fish), *Hippocampus* (sea horse; male has a brood pouch), *Echeneis* (sucker fish); Fresh water fishes: *Catla* (katla), *Labeo* (Rohu), *Clarias* (magur); Aquarium fishes: *Betta splendens* (Siamese fighting fish), *Pterophyllum* (angel fish).

"Teleostians are the most successful group of fishes". Explain why?

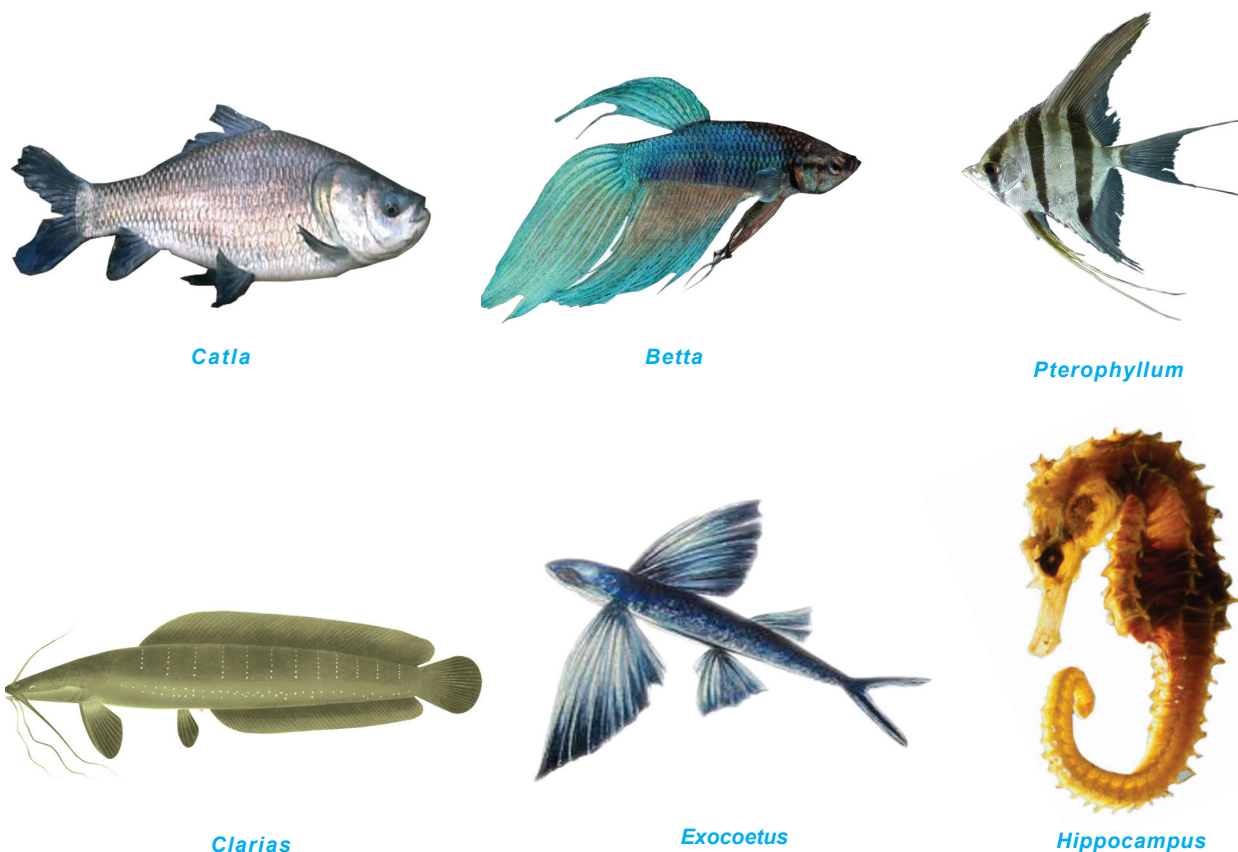


Fig. 4.11 Bony fishes